

TITLE PAGE

- **Problem Statement ID – 1**
- **Problem Statement Title- Eco-Loop Building Agents**
- **Theme- Clean & Green Technology**
- **PS Category- Software**
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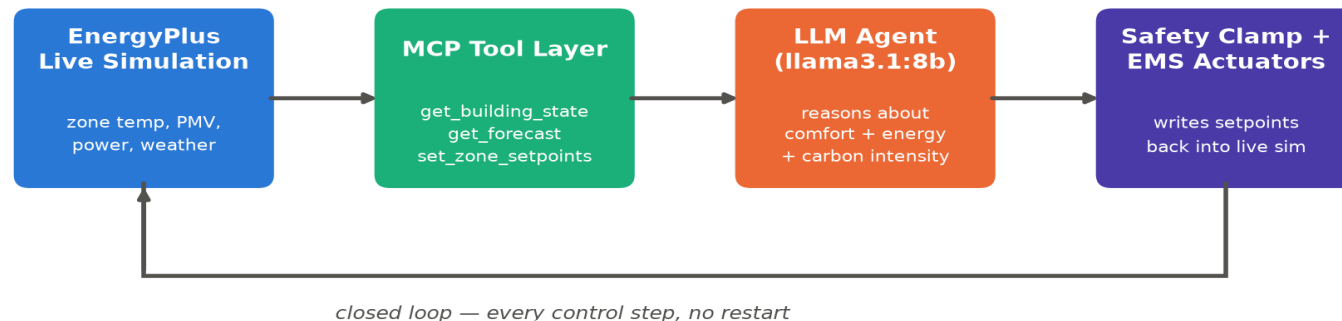
Eco-Loop Building Agents — Closed-Loop LLM Control for Building Energy

Proposed Solution (Describe your Idea/Solution/Prototype)

- A closed-loop AI that reads a live EnergyPlus simulation and writes real HVAC setpoint decisions back into it every control step - proven live, not just simulated on paper.
- Cuts building energy use while protecting occupant comfort, verified end-to-end with a real local LLM - not a rule engine or a spreadsheet estimate.
- Detailed explanation of the proposed solution
 - Live EnergyPlus + a local LLM agent (Llama 3.1 8B) that reads sensor state every step via a real MCP server
 - Agent decides setpoints -> safety-clamped -> written straight back into the running simulation (EMS actuators)
- How it addresses the problem
 - Rigid BMS schedules can't react to real-time weather, occupancy, or grid carbon intensity
 - **Verified: 0.9% energy reduction, comfort held at 91.9% (vs. 97.5% baseline)**
- Innovation and uniqueness of the solution
 - Genuine MCP tool-calling agent, not a rules engine relabeled "AI" - open-source, fully local
 - Safety-first: clamped decisions + auto-fallback -> 0 crashes across 80+ live decisions

TECHNICAL APPROACH

- Technologies to be used (e.g. programming languages, frameworks, hardware)
 - Python, EnergyPlus 26.1 (Python API / EMS), pandas, matplotlib
 - Ollama running Llama 3.1 8B (open-source LLM, fully local)
 - Model Context Protocol (MCP) for the agent's tool-calling layer
- Methodology and process for implementation (Flow Charts/Images/ working prototype)
 - Built and verified end-to-end - see closed-loop flow below

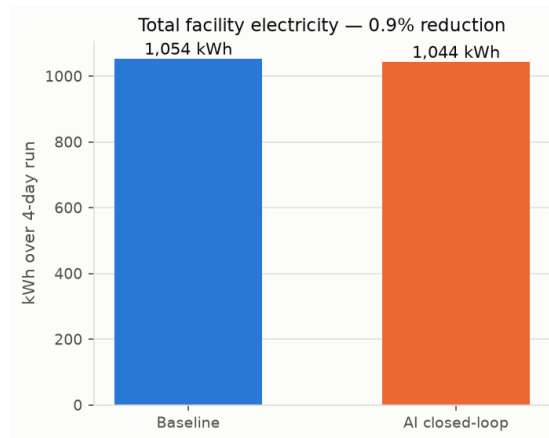


FEASIBILITY AND VIABILITY

- Analysis of the feasibility of the idea
 - Already built and verified end-to-end - real EnergyPlus + real LLM + real MCP, tested across multiple full runs
 - 100% open-source, runs on one commodity machine - no cloud spend
- Potential challenges and risks
 - LLM latency and variability (20-125s per decision on an 8B local model)
 - Small model doesn't always follow multi-step rules consistently
- Strategies for overcoming these challenges
 - Hard timeout + fallback to last-known-good setpoints -> 0 crashes across 80+ decisions
 - Iteratively strengthened prompts lifted comfort compliance from 79% to 91%

ARTIFACTS

- Relevant artifacts, such as :
 - Copy of the Code Embedded - full source at github.com/srivatsa927/Eco-Loop-Building-Agents
 - Snaps of the solution proposal
 - Dashboard snaps



Energy savings —
dashboard.html

```
src/safety.py

# Hard bounds applied regardless of what the LLM requests
ABSOLUTE_MIN_COOLING_C, ABSOLUTE_MAX_COOLING_C = 22.0, 29.0
ABSOLUTE_MIN_HEATING_C, ABSOLUTE_MAX_HEATING_C = 15.0, 22.0
MIN_DEADBAND_C = 1.5 # cooling must stay this far above heating

def clamp_setpoints(cooling_c, heating_c):
    cooling_c = max(MIN_COOLING, min(MAX_COOLING, cooling_c))
    heating_c = max(MIN_HEATING, min(MAX_HEATING, heating_c))
    if cooling_c - heating_c < MIN_DEADBAND_C:
        mid = (cooling_c + heating_c) / 2
        cooling_c, heating_c = mid + 0.75, mid - 0.75
    return {"cooling_c": cooling_c, "heating_c": heating_c}

# Shared by the MCP server AND the EnergyPlus runner - the model's
# raw output never reaches an actuator unclamped.
```

Safety-clamped setpoints — src/safety.py

RESEARCH AND REFERENCES

- Details / Links of the reference and research work
 - EnergyPlus & DOE reference building models - energyplus.net
 - Model Context Protocol specification - modelcontextprotocol.io
 - Meta Llama 3.1 (model card) served locally via Ollama - ollama.com
 - Prior art: EnergyPlus EMS Python API used for RL-based HVAC control research (e.g. RL-EmsPy, RL-BCA)
 - **Project repo: github.com/srivatsa927/Eco-Loop-Building-Agents**